

Sharp Gaussian Brenier stability under logarithmic second moments

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Abstract

A logarithmic strengthening of the target second moment determines an exact stability exponent for Brenier maps from a fixed Gaussian source. Under the normalized bound $\int (|y|/M)^2 [\log(e + (|y|/M)^2)]^\beta d\lambda \leq 1$, $\beta > 0$, we prove a homogeneous L^2 map estimate with Wasserstein exponent $\min\{1/3, \beta/(\beta + 1)\}$. This exponent is optimal in every fixed dimension at least two, including the transition $\beta = 1/2$ without a logarithmic loss. At $\beta = 0$ there is no uniform modulus in these dimensions. The proof combines the previously established radius-free potential estimate with a minimum-density finite-difference inequality. Integrating signed-square derivative increments directly against the common endpoint weight requires only L^2 gradients and produces polynomial source-coordinate weights. A dependence-free logarithmic-moment pairing controls these weights. We also prove uniform estimates for full-support strongly convex $C^{1,1}$ source potentials with a fixed gradient-Lipschitz bound. As a further consequence, the best Gaussian finite- q one-third constant, in its natural homogeneous normalization, has exact order $(q - 2)^{-1/6}$ as $q \downarrow 2$. The sharpness and endpoint obstructions are multidimensional; in one dimension the map distance equals W_2 .

Status and proof dependency. This is a research draft, not a formally verified or externally peer-reviewed paper. No publication-priority claim is made. The all- $\mathcal{P}_2$ potential theorem from the public version 3 manuscript [1] is an explicit input, restated below with its exact hypotheses. The present version gives complete proofs of the new interpolation, logarithmic-moment, sharpness and critical-constant statements; it does not independently re-prove the earlier cell-calculus theorem. Comparison with the full text of [5] remains unfinished.

1 The logarithmic threshold

Let γ_d denote standard Gaussian probability on $\mathbb{R}^d$. For an absolutely continuous source $\rho \in \mathcal{P}_2(\mathbb{R}^d)$ and a target $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, write T_μ for the unique quadratic optimal transport map from ρ to μ . It is the gradient of a convex Brenier potential [2]. Here $\mathcal{P}_q$ denotes probability measures with finite q th moment, and

$$W_2(\mu, \nu)^2 = \inf_{\pi \in \Pi(\mu, \nu)} \int |y - z|^2 d\pi(y, z).$$

The Euclidean norm is $|\cdot|$, all logarithms are natural, and $t_+ = \max(t, 0)$.

For $\beta \geq 0$, define $\Phi_\beta(s) = s[\log(e + s)]^\beta$ for $s \geq 0$. This function is increasing and is at least s . Our target condition, with common scale $M > 0$, is

$$\int \Phi_\beta \left(\frac{|y|^2}{M^2} \right) d\lambda(y) \leq 1 \quad (\lambda = \mu, \nu). \tag{1.1}$$

It implies both target second moments are at most M^2 .

Theorem 1.1 (Sharp Gaussian logarithmic-moment estimate). *Let $d \geq 1$, $\beta > 0$, and put*

$$\eta = \min\{1, 2\beta\}, \quad \alpha = \frac{\eta}{\eta + 2} = \min\left\{\frac{1}{3}, \frac{\beta}{\beta + 1}\right\}, \quad (1.2)$$

$$K_\eta = 6 \cdot 2^\eta + 2, \quad B_\beta = 2 + 8^\beta + \sqrt{2}(8\beta/e)^\beta, \quad (1.3)$$

$$C_{d,\beta} = (24d + 2dK_\eta B_\beta)^{1/2}.$$

If μ, ν satisfy (1.1), their Brenier maps from γ_d obey

$$\|T_\mu - T_\nu\|_{L^2(\gamma_d)} \leq C_{d,\beta} M^{1-\alpha} W_2(\mu, \nu)^\alpha. \quad (1.4)$$

Moments may instead be measured about any common center $z \in \mathbb{R}^d$.

For each fixed $d \geq 2$, the exponent α cannot be increased on the normalized class (1.1) with $M = 1$. At $\beta = 0$, that class has no uniform map-continuity modulus. In dimension one,

$$\|T_\mu - T_\nu\|_{L^2(\gamma_1)} = W_2(\mu, \nu), \quad (1.5)$$

including the second-moment endpoint.

The transition at $\beta = 1/2$ is included: its exponent is exactly $1/3$. The constants displayed here are admissible, not asserted optimal. For $\beta > 0$ the condition already implies uniform integrability of the second moments, since

$$\int_{|y|>R} |y|^2 d\lambda(y) \leq \frac{M^2}{[\log(e + (R/M)^2)]^\beta}.$$

Thus qualitative uniform continuity also follows from classical transport stability and W_2 compactness. The issue addressed here is the explicit sharp power law and its endpoint.

1.1 A smooth-source extension

The same exponent holds for full-support uniformly strongly log-concave sources with a controlled gradient of their logarithmic density.

Theorem 1.2 (Uniform smooth-source estimate). *Let $V : \mathbb{R}^d \rightarrow \mathbb{R}$ be continuously differentiable, κ -strongly convex, and have globally Λ -Lipschitz gradient, where $0 < \kappa \leq \Lambda < \infty$. Let $\rho = Z^{-1}e^{-V} dx$. For $\beta > 0$ set η, α, K_η as in (1.2)–(1.3) and define*

$$B_{d,\kappa,\Lambda,\beta} = 2 + (8/\kappa)^\beta + (8\beta/(\kappa e))^\beta (2\Lambda/\kappa)^{d/2}, \quad (1.6)$$

$$C_{d,\kappa,\Lambda,\beta} = \max \left\{ 2, \left(\frac{24d}{\kappa} + 2dK_\eta \Lambda^\eta B_{d,\kappa,\Lambda,\beta} \right)^{1/2} \right\}. \quad (1.7)$$

For targets satisfying (1.1),

$$\|T_\mu - T_\nu\|_{L^2(\rho)} \leq C_{d,\kappa,\Lambda,\beta} M^{1-\alpha} W_2(\mu, \nu)^\alpha. \quad (1.8)$$

The constant is uniform in V for fixed $d, \kappa, \Lambda, \beta$, and common-center moments may be used. If $d \geq 2$, the exponent is sharp over this source class; individual sharpness for every non-Gaussian member is not asserted. Hard convex boundaries are excluded from this theorem.

1.2 The critical finite-moment constant

The same interpolation improves the dependence on the target-moment order in the Gaussian one-third estimate.

Theorem 1.3 (Sharp finite- q constant order). *For $q > 2$, put*

$$p = \frac{q}{q-2}, \quad D_q = \|1 + |G|\|_{L^p(\gamma_1)}, \quad G \sim \gamma_1,$$

and, for $\mu, \nu \in \mathcal{P}_q(\mathbb{R}^d)$, let

$$M_q = \left(\int |y|^q d\mu(y) \right)^{1/q} + \left(\int |y|^q d\nu(y) \right)^{1/q}.$$

Then for the standard Gaussian source in every $d \geq 1$,

$$\|T_\mu - T_\nu\|_{L^2(\gamma_d)} \leq \sqrt{38d} D_q^{1/3} M_q^{2/3} W_2(\mu, \nu)^{1/3}. \quad (1.9)$$

Let $C_(d, q)$ be the smallest constant in the same inequality when the factor $\sqrt{38d} D_q^{1/3}$ is replaced by a constant uniform over all such targets. For $d \geq 2$, there are explicit $c_0 > 0$ and $A_d < \infty$, independent of q , with*

$$c_0 p^{1/6} \leq C_*(d, q) \leq A_d p^{1/6}. \quad (1.10)$$

In particular $C_(d, q) \asymp_d (q-2)^{-1/6}$ as $q \downarrow 2$. For $d = 1$, $C_*(1, q) = 1$.*

The last theorem concerns the best constant in the specified homogeneous normalization, rather than a different transport exponent. Explicit choices for c_0, A_d appear in Section 7.

1.3 Provenance and related estimates

The potential estimate used here is [1, Theorem 1.1]. That manuscript develops a strongly log-concave extension of the finite-cell and moving-site framework of the OpenAI uniform-source preprint [7]. The earlier Gaussian three-atom construction is used below with full details. The new interpolation keeps signed derivative increments under the minimum of the two endpoint densities; it does not estimate an exponential density ratio against the original weight.

Potential-to-gradient interpolation and unbounded-target moment estimates have important precedents. Delalande and Mériqot [3, Proposition 4.1 and Corollary 4.4] treat bounded-gradient interpolation and finite target moments for compact convex sources with two-sided density bounds. For moment order $p \geq 4$, $p > d$, their map exponent is $p/(6p + 16d)$ in W_1 . Letrouit and Mériqot [4, Theorem 1.4 and Remark 1.5] treat strongly log-concave sources and bounded logarithmic perturbations with compact targets; their unbounded-target case is left untreated there. Mischler and Trevisan [6] establish related potential and map estimates for general power costs and discuss weighted interpolation. These results have different source or target hypotheses. Neither a metric exponent nor the absence of a statement in the compared results is evidence of priority. A full comparison with [5] remains unfinished.

2 The inherited potential theorem

We state exactly the result from version 3 that is required below. Its full public proof, including nonsmooth source conditioning, finite-target approximation and completion to all $\mathcal{P}_2$ targets, is in the cited manuscript.

Theorem 2.1 (Potential stability input, [1, Theorem 1.1]). *Let $\rho = r \, dx$ be full-dimensional with*

$$r(x) = Z^{-1} \exp(-\kappa|x|^2/2 - W(x)), \quad \kappa > 0,$$

where $W : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ is proper, lower-semicontinuous and convex. For every $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, its Brenier potential has a unique centered representative $U_\mu \in L^2(\rho)$, finite and convex on the interior of the source support. For every $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\|U_\mu - U_\nu\|_{L^2(\rho)}^2 \leq \frac{2}{\kappa} W_2(\mu, \nu)^2. \quad (2.1)$$

This is an established input within the cited proof chain, not a new conditional hypothesis on the targets. For the Gaussian and smooth sources of the present paper the support is all of $\mathbb{R}^d$, so the representatives are finite convex functions everywhere. The Gaussian has $W = 0$, $\kappa = 1$. Under Theorem 1.2, $W = V - \kappa|x|^2/2$ is finite convex. The source has a finite second moment in both cases. No bounded-gradient conclusion is inferred for unbounded targets.

We also use the standard optimality of a convex gradient to its own law. For finite maxima of affine functions, it follows directly by integrating the supporting affine inequalities: the gradient plan maximizes $\int x \cdot y$, hence minimizes quadratic cost, and equality forces its unique maximizing slope almost everywhere. This verifies every explicit map in the sharpness examples below.

3 A minimum-density finite-difference inequality

The key point is to preserve the common endpoint weight when integrating the signed-square derivative increment. The following form does not assume log-concavity of the density.

Lemma 3.1 (General minimum-weight inequality). *Let $\rho = r \, dx$ be a probability on $\mathbb{R}^d$ with $0 < r < \infty$ almost everywhere. Let u, v be finite convex functions on $\mathbb{R}^d$, with gradients in $L^2(\rho)$ and $f = u - v - a \in L^2(\rho)$ for some $a \in \mathbb{R}$. For $h > 0$ and each coordinate j , define*

$$m_{j,h}(x) = \min\{r(x), r(x + he_j)\},$$

$$A_{j,h}(x) = \frac{|m_{j,h}(x - he_j) - m_{j,h}(x)|}{r(x)}, \quad B_{j,h}(x) = 1 - \frac{m_{j,h}(x)}{r(x)}.$$

Then $0 \leq A_{j,h}, B_{j,h} \leq 1$ almost everywhere and

$$\begin{aligned} \|\nabla u - \nabla v\|_{L^2(\rho)}^2 &\leq \frac{12d}{h^2} \|f\|_{L^2(\rho)}^2 \\ &\quad + \sum_{j=1}^d \int (6A_{j,h} + 2B_{j,h})(|\partial_j u|^2 + |\partial_j v|^2) \, d\rho. \end{aligned} \quad (3.1)$$

Proof. Fix j, h and abbreviate $m = m_{j,h}$. For this fixed displacement, discard the null differentiability sets and their translates. Convexity gives

$$\partial_j u(x) \leq D_h u(x) := \frac{u(x + he_j) - u(x)}{h} \leq \partial_j u(x + he_j).$$

Set $a_u = \partial_j u$, $g_u = a_u |a_u|$ and $\delta_u = D_h u - a_u \geq 0$. The elementary inequality

$$(b - a)^2 \leq 2(b|b| - a|a|) \quad (a \leq b) \quad (3.2)$$

follows by factoring the difference of squares when the signs agree, and by $(|a| + |b|)^2 \leq 2(a^2 + b^2)$ when they differ. Consequently

$$\delta_u(x)^2 \leq 2(g_u(x + he_j) - g_u(x)),$$

with a nonnegative bracket.

Both terms are absolutely integrable under the minimum weight using only the assumed L^2 gradient:

$$\begin{aligned} \int |g_u(x)|m(x) dx &\leq \int |a_u|^2 r dx < \infty, \\ \int |g_u(x + he_j)|m(x) dx &= \int |g_u(y)|m(y - he_j) dy \leq \int |a_u|^2 r dy < \infty. \end{aligned}$$

The second bound uses $m(y - he_j) = \min(r(y - he_j), r(y)) \leq r(y)$. Signed subtraction and a change of variable are therefore legitimate. They give

$$\begin{aligned} \int \delta_u^2 m dx &\leq 2 \int g_u(y)(m(y - he_j) - m(y)) dy \\ &\leq 2 \int |a_u|^2 A_{j,h} d\rho. \end{aligned} \tag{3.3}$$

The first signed integral is nonnegative by its preceding representation; its integrand after translation need not be pointwise nonnegative. Taking an absolute-value upper bound is valid. The same calculation holds for v .

The common weight controls both finite-difference endpoints:

$$\begin{aligned} \int |D_h f|^2 m dx &\leq \frac{2}{h^2} \int (|f(x + he_j)|^2 + |f(x)|^2) m(x) dx \\ &\leq \frac{4}{h^2} \|f\|_{L^2(\rho)}^2. \end{aligned} \tag{3.4}$$

No translated $L^2(\rho)$ assumption on f is used. Since $a_u - a_v = D_h f - \delta_u + \delta_v$, the three-term square inequality and (3.3)–(3.4) yield

$$\int |a_u - a_v|^2 m dx \leq \frac{12}{h^2} \|f\|_2^2 + 6 \int A_{j,h} (|a_u|^2 + |a_v|^2) d\rho.$$

On the remaining weight, $|a_u - a_v|^2 \leq 2(|a_u|^2 + |a_v|^2)$ gives the term with $2B_{j,h}$. Summing in j proves (3.1). The bounds $A_{j,h}, B_{j,h} \leq 1$ follow because both minimum densities in $A_{j,h}$ lie between zero and $r(x)$. $\square$

Corollary 3.2 (Gaussian polynomial-weight interpolation). *Under the assumptions of Lemma 3.1 with $\rho = \gamma_d$, for $0 < \eta \leq 1$ and $0 < h \leq 1$,*

$$\begin{aligned} \|\nabla u - \nabla v\|_2^2 &\leq 12dh^{-2} \|u - v - a\|_2^2 \\ &\quad + K_\eta h^\eta \sum_j \int (|\partial_j u|^2 + |\partial_j v|^2) (1 + |x_j|)^\eta d\gamma_d. \end{aligned} \tag{3.5}$$

The last integrals may initially be infinite; the inequality remains valid in the extended-real sense.

Proof. Put $F(s) = \min(1, e^s)$. This function takes values in $(0, 1]$ and is globally 1-Lipschitz. Gaussian translation gives

$$\frac{m_{j,h}(x - he_j) - m_{j,h}(x)}{r(x)} = F(hx_j - h^2/2) - F(-hx_j - h^2/2).$$

Hence $A_{j,h} \leq \min(2h|x_j|, 1)$. Also

$$B_{j,h} = (1 - e^{-hx_j - h^2/2})_+ \leq \min\{h(|x_j| + \tfrac{1}{2}), 1\}.$$

For $t \geq 0$ and $0 < \eta \leq 1$, $\min(t, 1) \leq t^\eta$. Therefore

$$6A_{j,h} + 2B_{j,h} \leq h^\eta (6 \cdot 2^\eta |x_j|^\eta + 2(|x_j| + \tfrac{1}{2})^\eta) \leq K_\eta h^\eta (1 + |x_j|)^\eta.$$

Insert this into (3.1). □

4 A dependence-free logarithmic pairing

The map coordinate and the source coordinate in the next application are generally dependent. No independence is needed.

Lemma 4.1 (Logarithmic moment pairing). *On a probability space, let $A, S \geq 0$ be arbitrary jointly distributed variables. Assume $\beta > 0$, $a > 0$ and*

$$\mathbb{E}\Phi_\beta(A) \leq 1, \quad \mathbb{E}e^{aS^2} \leq H < \infty.$$

If $0 < \eta \leq \min(1, 2\beta)$, then

$$\mathbb{E}[A(1 + S)^\eta] \leq 2 + (2/a)^\beta + (2\beta/(ae))^\beta H. \quad (4.1)$$

Proof. Since $\Phi_\beta(A) \geq A$, $\mathbb{E}A \leq 1$. For $s \geq 0$, $(1 + s)^\eta \leq 1 + s^\eta \leq 2 + s^{2\beta}$. On the event $S^2 \leq (2/a) \log(e + A)$,

$$AS^{2\beta} \leq (2/a)^\beta \Phi_\beta(A).$$

On the complementary event, $A < e^{aS^2/2}$, and

$$AS^{2\beta} \leq e^{aS^2/2} S^{2\beta} \leq (2\beta/(ae))^\beta e^{aS^2}.$$

The last inequality follows by maximizing $x^\beta e^{-ax/2}$ over $x \geq 0$, whose maximum is $(2\beta/(ae))^\beta$. Taking expectations proves (4.1). □

For $S = |G|$, $G \sim \gamma_1$, take $a = 1/4$ and $H = \sqrt{2}$, since $\mathbb{E}e^{G^2/4} = \sqrt{2}$. Then (4.1) gives exactly B_β in (1.3).

Upper estimate in Theorem 1.1. The logarithmic condition puts both targets in $\mathcal{P}_2$. Let $u = U_\mu$, $v = U_\nu$ be the centered potentials from Theorem 2.1, finite on all of $\mathbb{R}^d$. Set $w = W_2(\mu, \nu)$. Their difference satisfies $\|u - v\|_2^2 \leq 2w^2$. For each j , apply Lemma 4.1 with

$$A = |\partial_j u|^2 / M^2, \quad S = |x_j|.$$

The source coordinate is standard Gaussian. Monotonicity of Φ_β , the inequality $|\partial_j u| \leq |\nabla u|$, and $(\nabla u)_\# \gamma_d = \mu$ verify its logarithmic hypothesis. The same holds for v . Hence

$$\sum_j \int (|\partial_j u|^2 + |\partial_j v|^2) (1 + |x_j|)^\eta d\gamma_d \leq 2dB_\beta M^2.$$

By Corollary 3.2,

$$\|T_\mu - T_\nu\|_2^2 \leq 24dw^2h^{-2} + 2dK_\eta B_\beta M^2 h^\eta. \quad (4.2)$$

If $0 < w \leq M$, choose $h = (w/M)^{2/(\eta+2)} \in (0, 1]$. Both terms have scale $M^{4/(\eta+2)}w^{2\eta/(\eta+2)}$, and taking square roots proves (1.4). If $w > M$, the elementary bound $\|T_\mu - T_\nu\|_2 \leq 2M$ proves the same conclusion since $C_{d,\beta} \geq 2$. If $w = 0$, uniqueness gives equality of the maps. The scale M is strictly positive, so no division by zero is involved.

For a common center z , subtract $z \cdot x$ from both convex potentials. Their difference and the map difference remain unchanged, and the gradient laws are the common translates of the targets. The same argument therefore proves the centered-moment version. $\square$

5 Sharpness and the second-moment endpoint

We first establish a reusable two-atom calculation. Its radius will then be chosen to fit the logarithmic normalization or a fixed power moment.

Lemma 5.1 (Rare rotated halfspaces). *Let $d \geq 2$, $t \geq 1$, and let G, Z be independent standard normals. Put*

$$m_t = \Pr(G > t), \quad n_t = \cos(1/t)e_1 + \sin(1/t)e_2, \quad \delta_t = |e_1 - n_t|.$$

For any $R > 0$, define

$$\mu = (1 - m_t)\delta_0 + m_t\delta_{Re_1}, \quad \nu = (1 - m_t)\delta_0 + m_t\delta_{Rn_t}.$$

Their Gaussian Brenier maps are $Re_1\mathbf{1}_{A_t}$ and $Rn_t\mathbf{1}_{B_t}$, where $A_t = \{x_1 > t\}$ and $B_t = \{x \cdot n_t > t\}$. If $D_t = \gamma_d(A_t \triangle B_t)$, then

$$W_2(\mu, \nu)^2 = m_t R^2 \delta_t^2, \quad (5.1)$$

$$\|T_\mu - T_\nu\|_2^2 = R^2 [D_t + \delta_t^2(m_t - D_t/2)], \quad (5.2)$$

$$c m_t \leq D_t \leq 2m_t, \quad c = e^{-3/2} \Pr(Z < -2) > 0. \quad (5.3)$$

Also

$$e^{-3/2} \frac{\varphi(t)}{t} \leq m_t \leq \frac{\varphi(t)}{t}, \quad \delta_t = 2 \sin(1/(2t)) \sim t^{-1}, \quad (5.4)$$

where φ is the standard normal density.

Proof. The maps are gradients of $R(x_1 - t)_+$ and $R(x \cdot n_t - t)_+$. Rotational invariance gives both rare cells mass m_t ; convexity and the supporting-inequality argument identify their optimal maps. Since $m_t < 1/2$, a two-by-two target coupling is specified by the mass $s \in [0, m_t]$ sent between the two nonzero atoms. Its squared cost is

$$R^2(2(m_t - s) + s\delta_t^2) = 2R^2(m_t - s \cos(1/t)).$$

This is minimized at $s = m_t$ because $\cos(1/t) > 0$, proving (5.1). On the symmetric difference the squared map discrepancy is R^2 ; on the overlap it is $R^2\delta_t^2$. The overlap mass is $m_t - D_t/2$, giving (5.2).

The event $E_t = \{t < G < t + 1/t, Z < -2\}$ lies in $A_t \setminus B_t$. Indeed $\sin x \geq 5x/6$ for $0 < x \leq 1$, and on this event

$$G \cos(1/t) + Z \sin(1/t) < t + 1/t - 5/(3t) < t.$$

Monotonicity of φ on $[t, \infty)$ gives

$$\Pr(t < G < t + 1/t) \geq \frac{\varphi(t + 1/t)}{t} \geq e^{-3/2} \frac{\varphi(t)}{t}.$$

Furthermore

$$m_t \leq \int_t^\infty (s/t) \varphi(s) ds = \varphi(t)/t.$$

Independence of G, Z therefore gives $\Pr(E_t) \geq cm_t$, proving the lower bound in (5.3); its upper bound follows by the union bound. The same interval estimate is a lower bound for m_t and proves (5.4). The identity for δ_t follows from elementary trigonometry. $\square$

5.1 The logarithmic rare-event obstruction

For $\beta > 0$, choose $R_t > 0$ uniquely by

$$m_t R_t^2 [\log(e + R_t^2)]^\beta = 1. \quad (5.5)$$

The left side is continuous and strictly increasing in R_t from zero to infinity. The targets in Lemma 5.1 then satisfy (1.1) with $M = 1$ and equality.

We record the scale rigorously. Set $Q_t = R_t^2$ and $\ell_t = \log(1/m_t)$. From (5.4),

$$\ell_t = t^2/2 + \log t + O(1) \sim t^2/2. \quad (5.6)$$

Equation (5.5) first implies $Q_t \rightarrow \infty$, since a bounded subsequence would make its left side tend to zero. Taking logarithms gives

$$\ell_t = \log Q_t + \beta \log \log(e + Q_t).$$

The second term is $o(\log Q_t)$, so $\log(e + Q_t) \sim \ell_t$. Returning to (5.5) yields

$$m_t R_t^2 = [\log(e + R_t^2)]^{-\beta} \sim (t^2/2)^{-\beta}. \quad (5.7)$$

Write $w_t = W_2(\mu_t, \nu_t)$. The exact identities and bounds in Lemma 5.1 now imply

$$w_t \sim 2^{\beta/2} t^{-\beta-1}, \quad \|T_{\mu_t} - T_{\nu_t}\|_2 \asymp_\beta t^{-\beta}. \quad (5.8)$$

For the second assertion, (5.2) is bounded below by $cm_t R_t^2$ and above by $(2 + \delta_t^2)m_t R_t^2$. An exact asymptotic of D_t/m_t is unnecessary. Consequently, for every $\theta > \beta/(\beta + 1)$,

$$\frac{\|T_{\mu_t} - T_{\nu_t}\|_2}{w_t^\theta} \longrightarrow \infty.$$

This gives the matching obstruction for $0 < \beta \leq 1/2$.

At $\beta = 0$, take $R_t = m_t^{-1/2}$. Both second moments are exactly one, while (5.1) gives $w_t = \delta_t \rightarrow 0$ and (5.2)–(5.3) give a uniform positive lower bound for map distance. This excludes every uniform modulus at the endpoint, not just positive Hölder exponents.

5.2 The bounded one-third obstruction

For completeness we reconstruct the Gaussian three-atom family from [1, Theorem 1.3], which adapts the three-plane mechanism in [7]. Let

$$F(s) = \Pr(|G| < s) \quad (s \geq 0), \quad b = ka,$$

where $k > 0$ is fixed and $a \downarrow 0$. There is a unique $c_a \in \mathbb{R}$ satisfying

$$\mathbb{E}F((c_a + kaZ)_+) = F(a) =: m_a. \quad (5.9)$$

The expectation is continuous, strictly increasing in c_a , and ranges from zero to one. Define

$$u_a(x) = \max\{-x_1, x_1, a\}, \quad v_a(x) = \max\{-x_1, x_1, c_a + kax_2\}.$$

Their central cells are $A_a = \{|x_1| < a\}$ and $B_a = \{|x_1| < (c_a + kax_2)_+\}$. They have equal mass by (5.9). Symmetry in x_1 gives the induced targets

$$\begin{aligned} \mu_a &= \frac{1}{2}(1 - m_a)(\delta_{-e_1} + \delta_{e_1}) + m_a\delta_0, \\ \nu_a &= \frac{1}{2}(1 - m_a)(\delta_{-e_1} + \delta_{e_1}) + m_a\delta_{be_2}. \end{aligned}$$

Their convex gradients are the optimal maps. Matching the two outer atoms and moving the central atom costs m_ab^2 , and every coupling has at least that second-coordinate cost. Hence

$$W_2(\mu_a, \nu_a)^2 = m_ab^2. \quad (5.10)$$

Let c_* be the unique real number with $\mathbb{E}(c_* + kZ)_+ = 1$. Since $F(s) \leq 2\varphi(0)s$ and $F(as)/a \rightarrow 2\varphi(0)s$, dominated convergence gives, for each fixed c ,

$$a^{-1}\mathbb{E}F(a(c + kZ)_+) \rightarrow 2\varphi(0)\mathbb{E}(c + kZ)_+.$$

Monotone bracketing at fixed $c_- < c_* < c_+$ in (5.9), followed by arbitrarily close brackets, proves $c_a/a \rightarrow c_*$. In particular these ratios are bounded. The two central intervals are nested on every x_2 slice, so

$$D_a := \gamma_d(A_a \triangle B_a) = \mathbb{E}|F((c_a + kaZ)_+) - F(a)|.$$

The integrand divided by a is dominated by a fixed multiple of $1 + |Z|$. Thus

$$D_a/a \rightarrow 2\varphi(0)\mathbb{E}|(c_* + kZ)_+ - 1| > 0.$$

The limit is positive because $(c_* + kZ)_+$ is not constant. Coordinate orthogonality gives exactly

$$|T_{\mu_a} - T_{\nu_a}|^2 = \mathbf{1}_{A_a \triangle B_a} + b^2\mathbf{1}_{B_a}, \quad \|T_{\mu_a} - T_{\nu_a}\|_2^2 = D_a + m_ab^2.$$

Since $m_a \sim 2\varphi(0)a$, the target distance is of order $a^{3/2}$ and the map distance of order $a^{1/2}$. All targets lie in B_1 for small a .

To fit any fixed logarithmic class with $M = 1$, multiply every atom by

$$s_\beta = [\log(e + 1)]^{-\beta/2} \leq 1.$$

On the scaled support, $|y|^2[\log(e + |y|^2)]^\beta \leq s_\beta^2[\log(e + 1)]^\beta = 1$. Scaling the convex potentials scales the maps, so both map distance and W_2 are multiplied by s_β . Every exponent above $1/3$ still fails. This is the matching obstruction for $\beta \geq 1/2$; at the transition both constructions match.

Together with the upper proof, this completes Theorem 1.1 in $d \geq 2$. All examples use only two source coordinates. In $d = 1$, the two increasing Brenier maps use a common source-quantile parameter, so their joint law is the monotone optimal coupling of the targets. This proves (1.5) and its stated exceptional scope.

6 Uniform full-support smooth sources

Proof of Theorem 1.2. Strong convexity gives a unique minimizer x_* of V with $\nabla V(x_*) = 0$. Put $S(x) = |x - x_*|$. Strong convexity and integration of the gradient-Lipschitz bound give

$$|\nabla V(x)| \leq \Lambda S(x), \quad V(x_*) + \frac{\kappa}{2} S(x)^2 \leq V(x) \leq V(x_*) + \frac{\Lambda}{2} S(x)^2.$$

Therefore $0 < Z < \infty$ and

$$r(x) \leq (\Lambda/(2\pi))^{d/2} e^{-\kappa S(x)^2/2}, \quad \mathbb{E}_\rho e^{\kappa S^2/4} \leq (2\Lambda/\kappa)^{d/2}. \quad (6.1)$$

The normalization bound uses $Z \geq e^{-V(x_*)} (2\pi/\Lambda)^{d/2}$; the exponential-moment bound is the integral of the resulting Gaussian envelope.

Use the same $F(s) = \min(1, e^s)$ as above. The difference of minimum weights satisfies

$$\frac{m_{j,h}(x - he_j) - m_{j,h}(x)}{r(x)} = F(V(x) - V(x - he_j)) - F(V(x) - V(x + he_j)).$$

Because F is 1-Lipschitz with unit range, and $0 < h \leq 1$,

$$\begin{aligned} A_{j,h}(x) &\leq \min\{|V(x + he_j) - V(x - he_j)|, 1\} \leq \min\{2h\Lambda(S(x) + 1), 1\}, \\ B_{j,h}(x) &\leq \min\{|V(x + he_j) - V(x)|, 1\} \leq \min\{h\Lambda(S(x) + 1), 1\}. \end{aligned}$$

Indeed integration on the segment bounds the first potential difference by $2h\Lambda S + \Lambda h^2$, and the second by $h\Lambda S + \Lambda h^2/2$. Thus Lemma 3.1 gives

$$\begin{aligned} \|\nabla u - \nabla v\|_{L^2(\rho)}^2 &\leq 12dh^{-2} \|u - v - a\|_{L^2(\rho)}^2 \\ &\quad + K_\eta \Lambda^\eta h^\eta \sum_j \int (|\partial_j u|^2 + |\partial_j v|^2) (1 + S)^\eta d\rho. \end{aligned} \quad (6.2)$$

Every signed-integral justification is already contained in the general minimum-weight lemma and uses only the original L^2 gradients.

For the centered Brenier potentials, apply Lemma 4.1 with $A = |\partial_j u|^2/M^2$, the radial variable S , $a = \kappa/4$, and $H = (2\Lambda/\kappa)^{d/2}$. The pushforward identity and (1.1) verify its target hypothesis, while (6.1) verifies its source hypothesis. Its constant is precisely (1.6). Hence the sum of the weighted gradient integrals in (6.2) is at most $2dB_{d,\kappa,\Lambda,\beta}M^2$.

Since $W = V - \kappa|x|^2/2$ is finite convex, Theorem 2.1 gives $\|u - v\|_2^2 \leq (2/\kappa)w^2$. We obtain

$$\|T_\mu - T_\nu\|_2^2 \leq (24d/\kappa)w^2h^{-2} + 2dK_\eta\Lambda^\eta B_{d,\kappa,\Lambda,\beta}M^2h^\eta.$$

For $0 < w \leq M$, take $h = (w/M)^{2/(\eta+2)}$. For $w > M$ use $\|T_\mu - T_\nu\|_2 \leq 2M$, and for $w = 0$ use uniqueness. This proves (1.8) with (1.7). Subtracting a common affine function gives the common-center statement.

For sharpness over the class, it contains the source $N(0, \kappa^{-1}I_d)$, whose gradient-Lipschitz constant is $\kappa \leq \Lambda$. If u is a potential for any standard-Gaussian example above, then $u(\sqrt{\kappa}x)/\sqrt{\kappa}$ has the same gradient law and map-discrepancy distribution under this dilated source. The target laws, logarithmic moments and Wasserstein distances are unchanged. The Gaussian obstructions therefore establish sharpness for the class in $d \geq 2$. Individual sharpness for non-Gaussian members is not claimed. $\square$

7 The sharp critical order of finite-moment constants

Proof of Theorem 1.3. Set $\eta = 1$ in (3.5), so $K_1 = 14$. Hölder's inequality with conjugate exponents $q/2$ and $p = q/(q-2)$ gives

$$\int |\partial_j u|^2 (1 + |x_j|) d\gamma_d \leq D_q \|\partial_j u\|_{L^q(\gamma_d)}^2.$$

For centered Brenier potentials, Theorem 2.1 and the gradient laws imply

$$\|T_\mu - T_\nu\|_2^2 \leq 24dw^2h^{-2} + 14dD_qM_q^2h. \quad (7.1)$$

Here the sum of coordinate L^q norms squared is at most d times the sum of the two vector L^q norms squared, hence at most dM_q^2 . If $w > 0$, then $M_q > 0$, and $w \leq M_q$ by the triangle inequality through δ_0 and L^q domination of L^2 . Since $D_q \geq 1$, the choice

$$h = \left(\frac{w}{M_q \sqrt{D_q}} \right)^{2/3} \quad (7.2)$$

lies in $(0, 1]$. Substitution in (7.1) gives coefficients $24d$ and $14d$ multiplying the common scale $D_q^{2/3} M_q^{4/3} w^{2/3}$. This proves (1.9). The zero cases follow from uniqueness; $M_q = 0$ means both targets are δ_0 .

For every $p \geq 1$, maximizing $s^p e^{-s^2/4}$ gives

$$\mathbb{E}|G|^p \leq (2p/e)^{p/2} \mathbb{E}e^{G^2/4} = \sqrt{2}(2p/e)^{p/2}.$$

Thus

$$D_q \leq 1 + 2\sqrt{p/e} \leq (1 + 2/\sqrt{e})\sqrt{p},$$

and the upper bound in (1.10) holds with

$$A_d = \sqrt{38d}(1 + 2/\sqrt{e})^{1/3}. \quad (7.3)$$

Retaining D_q in the admissible scale (7.2) is essential to this dependence; the scale need not be the exact numerical minimizer.

For the matching lower bound, fix $d \geq 2$, $q > 2$, and use Lemma 5.1 with any $t \geq 1$ and radius $R = m_t^{-1/q}$. Both q th moments are exactly one, so $M_q = 2$. Since $1 - 2/q = 1/p$, equations (5.1)–(5.3) imply

$$\frac{\|T_\mu - T_\nu\|_2}{M_q^{2/3} W_2(\mu, \nu)^{1/3}} \geq 2^{-2/3} \sqrt{c} m_t^{1/(3p)} \delta_t^{-1/3}. \quad (7.4)$$

Take $t = \sqrt{p} \geq 1$. The lower bound in (5.4) gives

$$m_t \geq e^{-3/2} (2\pi)^{-1/2} e^{-p/2} p^{-1/2}.$$

Therefore, using $(\log p)/p \leq 1/e$,

$$\begin{aligned} m_t^{1/(3p)} &\geq \exp \left(-\frac{1}{6} - \frac{1}{2p} - \frac{\log(2\pi)}{6p} - \frac{\log p}{6p} \right) \\ &\geq \exp \left(-\frac{1}{6} - \frac{1}{2} - \frac{\log(2\pi)}{6} - \frac{1}{6e} \right). \end{aligned}$$

Also $\delta_t = 2 \sin(1/(2t)) \leq 1/t = p^{-1/2}$. Inserting these bounds into (7.4) proves the lower inequality in (1.10) with

$$c_0 = 2^{-2/3} \sqrt{e^{-3/2} \Pr(G < -2)} \exp\left(-\frac{1}{6} - \frac{1}{2} - \frac{\log(2\pi)}{6} - \frac{1}{6e}\right) > 0. \quad (7.5)$$

This holds for every $p > 1$, not just asymptotically, and the source is fixed throughout. Since $p \sim 2/(q-2)$ as $q \downarrow 2$, the critical order follows.

In $d = 1$, the quantile identity gives map distance w , and $w \leq M_q$ implies $w \leq M_q^{2/3} w^{1/3}$. Two opposite nonzero Dirac targets have $w = M_q$, so equality is attainable and $C_*(1, q) = 1$. $\square$

8 Scope and relation to the earlier manuscript

As an integrability requirement, the logarithmic condition is weaker than any fixed higher-power moment condition. Its positive exponent is nevertheless possible because the minimum-density identity produces a polynomial Gaussian-coordinate weight, and the logarithmic target moment pairs with Gaussian exponential-square integrability. Neither that pairing nor the proof assumes independence between transport values and source points.

Version 3 [1] contains additional results for compact log-concave sources and hard-boundary target-tail estimates. They remain separate, established parts of that earlier manuscript. The present focused version does not extend the logarithmic theorem to arbitrary strongly log-concave sources with hard boundaries, nor to arbitrary bounded non-log-concave perturbations. The smooth-source theorem requires full support and the stated global gradient-Lipschitz bound. Sharpness for that class follows through its Gaussian member; no claim is made for every individual non-Gaussian source.

The sharpness of the exponent in Theorem 1.1, the absence of a modulus at $\beta = 0$, and the critical order in Theorem 1.3 concern fixed source dimension $d \geq 2$. Dimension one is governed by the exact quantile isometry. Mathematical proof and bibliographic priority are distinct issues; the latter is not asserted here.

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